

TABLE I
DIFFERENT DATA BITS

Sensor	PIR sensor output connected to pin of HT12E	Transmitted data bits (D3 D2 D1 D0)
Front sensor	AD8 (10)	1 1 1 0
Back sensor	AD9 (11)	1 1 0 1
Left sensor	AD10 (12)	1 0 1 1
Right sensor	AD11 (13)	0 1 1 1

through diodes D1 and D2. When $\overline{TE}$ pin is pulled low, IC1 transmits address A0 through A7, and AD8 through AD11 serially through RF Tx module. Because pin AD8 is pulled low, data bits are transmitted as 1110 (AD11 through AD8). Similarly, in other such circuits, when motion is detected, the respective data pin is pulled low, so different data is transmitted, as listed in Table I.

Thus, when a sensor detects motion from its side, different bit pattern of D0 through D3 is transmitted. At receiver side, this pattern is used to identify the side where motion is detected.

Receiver unit. As shown in Fig. 4, the receiver is built using 434MHz RF Rx module (RX1), RF decoder chip HT12D (IC5), AT89S52 MCU (IC6) and NE555 multivibrators (IC7 and IC8).

Serial data output from 434 RF Rx module is given to data input pin 14 of HT12D decoder.

Address pins A0 through A7 of HT12D are connected to ground to set same address 00 (0000 0000b) as encoder. LED5 is connected to VT (valid transmission) pin, which blinks to indicate that data is received. The same VT output is inverted using T5 and given to external interrupt pin EX1 (pin 13) of AT89S52. Data output pins D8 through D11 are connected to port pins P1.0 through P1.3, respectively.

LCD1 data pins D0 through D7 are connected to port P0 of IC6. Control pins RS and EN are connected to port pins P2.0 and P2.1 of IC6, respectively. Pin R/W is connected to ground to enable LCD1 write. Preset VR1 is connected to pin 3 to vary LCD1 brightness. Backlight LED pins 15 and 16 are connected to +5V via R23 and ground, respectively, to turn on LCD1 backlight.

There are two multivibrator circuits. Both are built using IC NE555 and configured in astable mode. Reset pin 4 of both multivibrators are controlled by MCU port pins P2.2 and P2.3.

A 12MHz crystal is connected to crystal

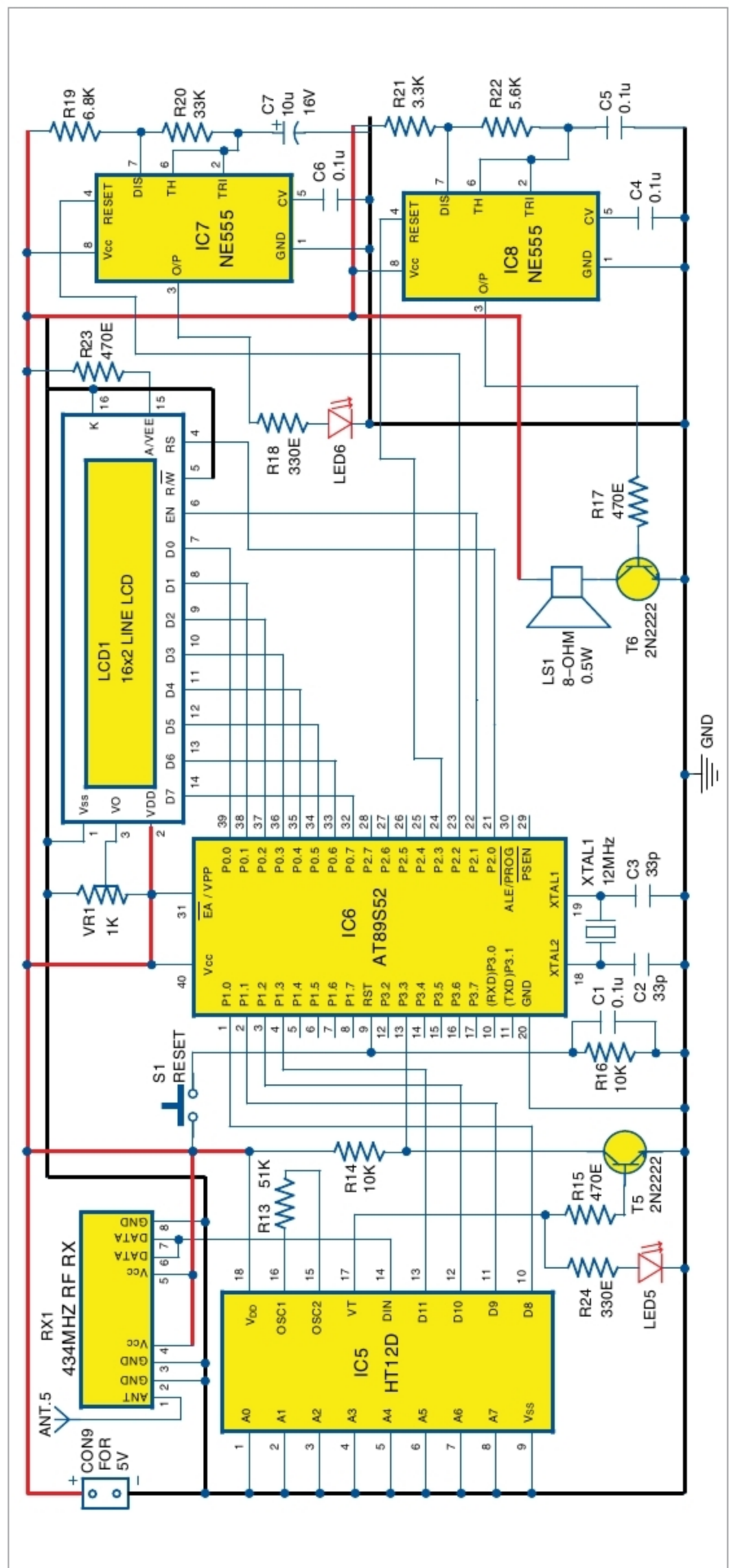


Fig. 4: Circuit diagram of MCU-based central receiver circuit